The Hyrule island 2

> Standard API: C++, C, Java, Python3

Link has a challenge at the Hyrule island. The challenge is visiting all shrines in this island. The island can be described by an $R \times C$ rectangle table. Each cell of the table is either a mountain, a shrine or an empty plot. Shrines are numbered from 1 to N. These mountains are so high and dangerous that Link can not enter these cells, Link can only enter shrines or empty plots.

Link must start at the shrine S, visit to all shrines and finish at the shrine F. The shrine S and the shrine F are different shrines. If he is in the cell (i, j), he can move to 8 cells next to this cell and not allowed to leave the island. Each such move takes 1 unit of time.

To win the challenge, Link must start at the shrine S, visit to all shrines and finish at the shrine F in the minimum time. Given the map of the Hyrule island, please help Link to find this minimum time.

Input Format

- The first line contains five positive integers $R, C, N, S, F (1 <= R, C <= 100; 2 <= N <= 9; 1 <= S, F <= N; S \neq F)$.
- In the next R lines, each line contains C characters consists of characters `.`, `#`, or an integer x $(1 <= x <= N)$ - that is an empty plot, a moutain or a shrine respectively. It is guaranteed that the map consists of N integers from 1 to N, each integer appears exactly once.

Output Format

- Ouput an integer which is the minimum time to visit all N shrines in the island, start at the shrine S and finish at the shrine F. If no such path exists, output -1.

Sample Input 1

```
4 6 4 1 4
#1..#.
#...2#
..#.#4
3.#...
```

Sample Output 1

```
8
```

Sample Input 2

```
4 6 4 4 3
#1..#.
#...2#
..#.#4
3.#...
```

Sample Output 2

```
7
```

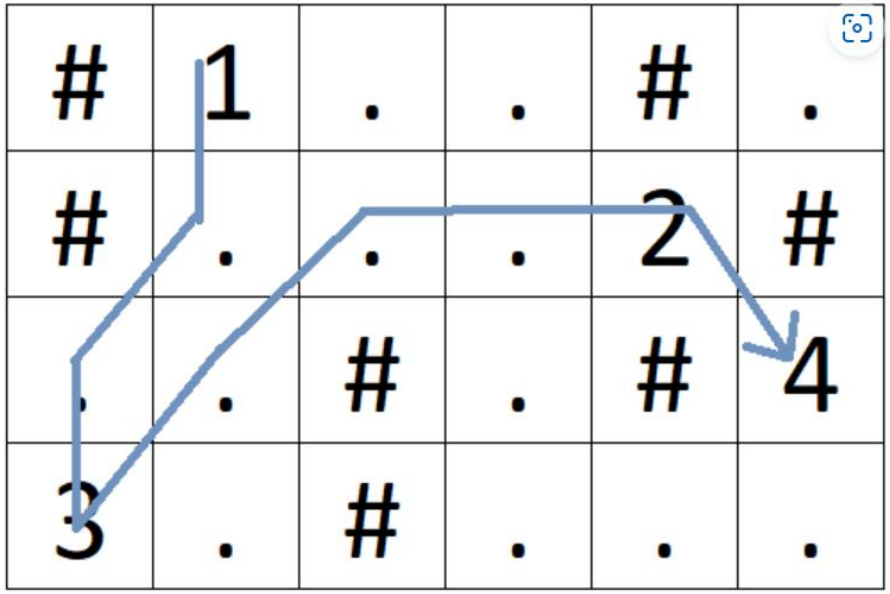
Sample Input 3

```
3 3 2 1 2
1##
###
##2
```

Sample Output 3

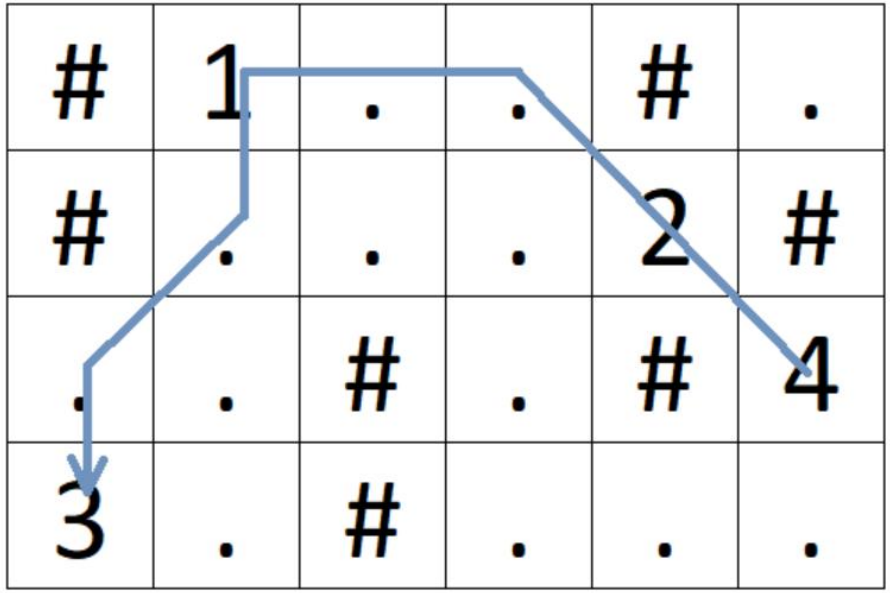
```
-1
```

Sample test 1:



In above image, Link started at shrine 1 and finished at shrine 4, he visited all shrines. With this path, Link made 8 moves, corresponds to 8 time units. This is the minimum time.

Sample test 2:



In above image, Link started at shrine 4 and finished at shrine 3, he visited all shrines. With this path, Link made 7 moves, corresponds to 7 time units. This is the minimum time.